

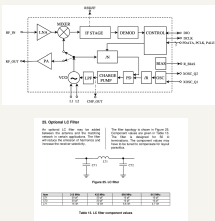
CC1000 All in one Transceiver 300MHz to 1 GHz
 315 / 433 / 868 and 915 Mhz

- High sensitivity (typical -110 dBm at 2.4kbaud)
- Programmable output power -20 to 10 dBm
- Low supply voltage (2.1 V to 3.6 V) MAX 5V

A Idela é implementar para as 3 frequências - 315 / 433 / 868
 O MCU controla qual com Switches?

Usar o RSSI para adaptar o ganho?

Pin	Library	Pin name	Pin type	Description
1	CC1000	RF_IN	RF	RF input pin
2	CC1000	RF_OUT	RF	RF output pin
3	CC1000	RF_IN	RF	RF input pin
4	CC1000	RF_OUT	RF	RF output pin
5	CC1000	RF_IN	RF	RF input pin
6	CC1000	RF_OUT	RF	RF output pin
7	CC1000	RF_IN	RF	RF input pin
8	CC1000	RF_OUT	RF	RF output pin
9	CC1000	RF_IN	RF	RF input pin
10	CC1000	RF_OUT	RF	RF output pin
11	CC1000	RF_IN	RF	RF input pin
12	CC1000	RF_OUT	RF	RF output pin
13	CC1000	RF_IN	RF	RF input pin
14	CC1000	RF_OUT	RF	RF output pin
15	CC1000	RF_IN	RF	RF input pin
16	CC1000	RF_OUT	RF	RF output pin
17	CC1000	RF_IN	RF	RF input pin
18	CC1000	RF_OUT	RF	RF output pin
19	CC1000	RF_IN	RF	RF input pin
20	CC1000	RF_OUT	RF	RF output pin
21	CC1000	RF_IN	RF	RF input pin
22	CC1000	RF_OUT	RF	RF output pin
23	CC1000	RF_IN	RF	RF input pin
24	CC1000	RF_OUT	RF	RF output pin
25	CC1000	RF_IN	RF	RF input pin
26	CC1000	RF_OUT	RF	RF output pin
27	CC1000	RF_IN	RF	RF input pin
28	CC1000	RF_OUT	RF	RF output pin
29	CC1000	RF_IN	RF	RF input pin
30	CC1000	RF_OUT	RF	RF output pin
31	CC1000	RF_IN	RF	RF input pin
32	CC1000	RF_OUT	RF	RF output pin
33	CC1000	RF_IN	RF	RF input pin
34	CC1000	RF_OUT	RF	RF output pin
35	CC1000	RF_IN	RF	RF input pin
36	CC1000	RF_OUT	RF	RF output pin
37	CC1000	RF_IN	RF	RF input pin
38	CC1000	RF_OUT	RF	RF output pin
39	CC1000	RF_IN	RF	RF input pin
40	CC1000	RF_OUT	RF	RF output pin



6.1 Input/output matching
 The CC1000 is the input match for the external L-match network. The external L-match network is used to match the transceiver to 50 Ω. An external L-match network is also available to connect the RF input and output together and match the transceiver to 50 Ω. For details, see the CC1000 datasheet.

6.2 VCO inductor
 The VCO is completely integrated except for the inductor L1.

6.3 Power supply decoupling
 The CC1000 is a low-power device and the power supply decoupling is not critical. The placement and size of the decoupling capacitors and the power supply voltage are not critical to achieve the optimum performance. Check the power supply voltage of the CC1000 and CC1000-200 that operate between 2.1V and 3.6V.

16. VCO
 Only one external inductor (L101) is required for the VCO. The inductor will determine the operating frequency range of the circuit. It is important to place the inductor as close to the pins as possible in order to reduce stray inductance. It is recommended to use a high Q, low tolerance inductor for best performance.

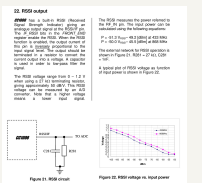
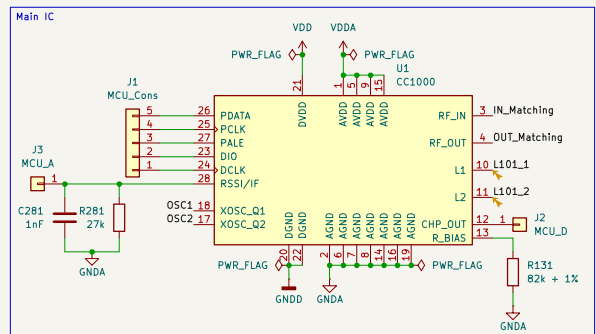
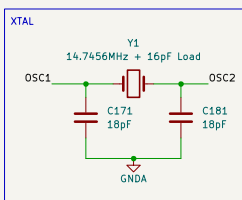
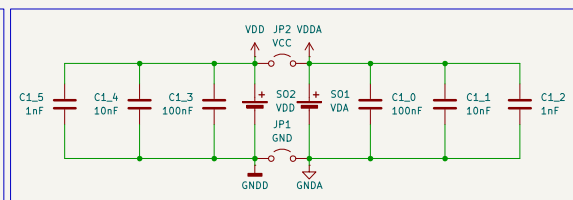


Table 1. Bill of materials for the application circuit

Item	315 MHz	433 MHz	868 MHz	915 MHz
1	1x 100 nF	1x 100 nF	1x 100 nF	1x 100 nF
2	1x 100 nF	1x 100 nF	1x 100 nF	1x 100 nF
3	1x 100 nF	1x 100 nF	1x 100 nF	1x 100 nF
4	1x 100 nF	1x 100 nF	1x 100 nF	1x 100 nF
5	1x 100 nF	1x 100 nF	1x 100 nF	1x 100 nF
6	1x 100 nF	1x 100 nF	1x 100 nF	1x 100 nF
7	1x 100 nF	1x 100 nF	1x 100 nF	1x 100 nF
8	1x 100 nF	1x 100 nF	1x 100 nF	1x 100 nF
9	1x 100 nF	1x 100 nF	1x 100 nF	1x 100 nF
10	1x 100 nF	1x 100 nF	1x 100 nF	1x 100 nF
11	1x 100 nF	1x 100 nF	1x 100 nF	1x 100 nF
12	1x 100 nF	1x 100 nF	1x 100 nF	1x 100 nF
13	1x 100 nF	1x 100 nF	1x 100 nF	1x 100 nF
14	1x 100 nF	1x 100 nF	1x 100 nF	1x 100 nF
15	1x 100 nF	1x 100 nF	1x 100 nF	1x 100 nF
16	1x 100 nF	1x 100 nF	1x 100 nF	1x 100 nF
17	1x 100 nF	1x 100 nF	1x 100 nF	1x 100 nF
18	1x 100 nF	1x 100 nF	1x 100 nF	1x 100 nF
19	1x 100 nF	1x 100 nF	1x 100 nF	1x 100 nF
20	1x 100 nF	1x 100 nF	1x 100 nF	1x 100 nF
21	1x 100 nF	1x 100 nF	1x 100 nF	1x 100 nF
22	1x 100 nF	1x 100 nF	1x 100 nF	1x 100 nF
23	1x 100 nF	1x 100 nF	1x 100 nF	1x 100 nF
24	1x 100 nF	1x 100 nF	1x 100 nF	1x 100 nF
25	1x 100 nF	1x 100 nF	1x 100 nF	1x 100 nF
26	1x 100 nF	1x 100 nF	1x 100 nF	1x 100 nF
27	1x 100 nF	1x 100 nF	1x 100 nF	1x 100 nF
28	1x 100 nF	1x 100 nF	1x 100 nF	1x 100 nF
29	1x 100 nF	1x 100 nF	1x 100 nF	1x 100 nF
30	1x 100 nF	1x 100 nF	1x 100 nF	1x 100 nF
31	1x 100 nF	1x 100 nF	1x 100 nF	1x 100 nF
32	1x 100 nF	1x 100 nF	1x 100 nF	1x 100 nF
33	1x 100 nF	1x 100 nF	1x 100 nF	1x 100 nF
34	1x 100 nF	1x 100 nF	1x 100 nF	1x 100 nF
35	1x 100 nF	1x 100 nF	1x 100 nF	1x 100 nF
36	1x 100 nF	1x 100 nF	1x 100 nF	1x 100 nF
37	1x 100 nF	1x 100 nF	1x 100 nF	1x 100 nF
38	1x 100 nF	1x 100 nF	1x 100 nF	1x 100 nF
39	1x 100 nF	1x 100 nF	1x 100 nF	1x 100 nF
40	1x 100 nF	1x 100 nF	1x 100 nF	1x 100 nF

Matching muda de frequência para frequência
 L101 muda indutor do VCO!

